

11 SUPERPOSITION

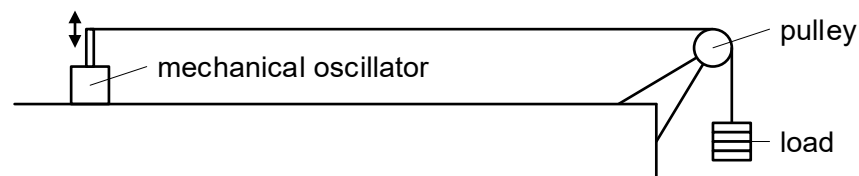
H2 Physics 9478



1.1 Discussion Questions

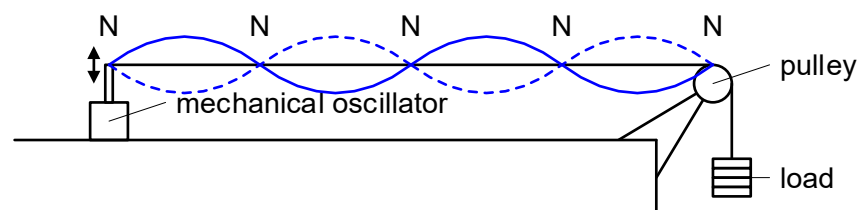
❖ Standing Waves

- D1** One end of a horizontal string is attached to a mechanical oscillator. The string passes over a pulley and the string is kept under tension by means of a weight as shown below.



The frequency of oscillation of the oscillator is increased and at certain frequencies, stationary waves are produced on the string.

- (a) Draw the stationary wave on the string when the frequency is such that the distance between the oscillator and the pulley corresponds to two wavelengths of the wave on the string. [1]
- (b) On your diagram, label the positions of the nodes N on the string. [1]



- (c) Explain why a stationary wave is observed on the string only at particular frequencies of vibration of the oscillator. [2]

The **speed** of the progressive wave travelling on the string is **constant** as the wire is uniform and the tension of the wire is constant. [1]

As the ends of the string are **nodes**, stationary waves can only exist (resonance occurs) when the length of the string is **integer multiple of $\frac{1}{2}$ -wavelength** of the progressive wave, i.e., there are **only certain wavelengths allowed**. [1]

Each of these wavelengths corresponds to one frequency of vibration. Hence only certain frequencies of vibration will produce stationary wave on the string.

[Speed of wave on string $v = \sqrt{T/\mu}$ where μ is the mass per unit length of the string.]

[J99/3/4 part]

- D2** A 60.0 cm long, vertical hollow tube is initially submerged just below the surface of water. As it is raised, the tube first resonates to a vibrating tuning-fork of frequency 512 Hz when the air column above the water level is 14.8 cm. The next resonance is observed when the length of air column is 48.0 cm. Determine

(a) the end-correction, and

[2]

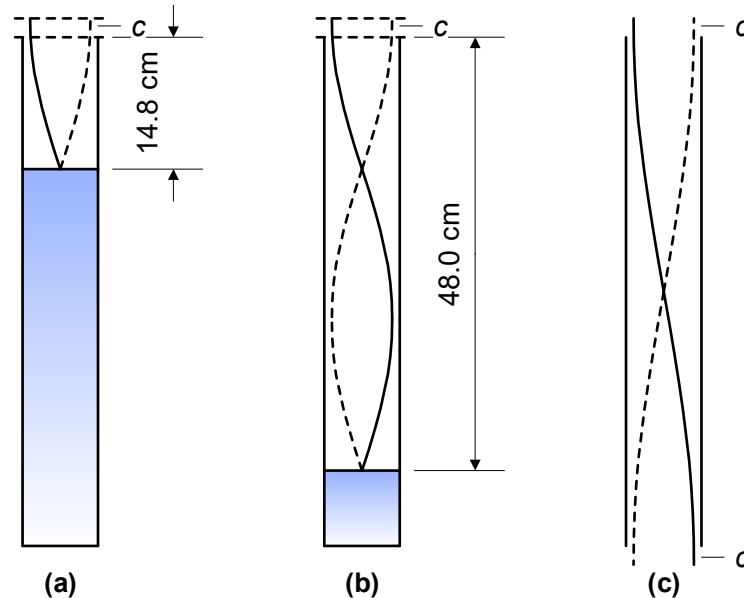


Fig. (a) and (b) shows the first two positions of the water level when resonance occurs.

(a): $\frac{1}{4} \lambda_{\text{closed}} = 14.8 + c$ ❶

(b): $\frac{3}{4} \lambda_{\text{closed}} = 48.0 + c$ ❷ [1]

❷ – 3 × ❶:

$0 = 3.6 - 2c$ or $c = 1.8 \text{ cm}$ [1]

- (b) the lowest frequency at which the tube will resonate when it is open at both ends. [3]

For the open pipe,

$\frac{1}{2} \lambda_{\text{open}} = 60.0 + 2c = 63.6 \text{ cm}$

$\lambda_{\text{open}} = 127.2 \text{ cm}$ [1]

Since $v = f \lambda$,

$\frac{f_{\text{open}}}{f_{\text{closed}}} = \frac{\lambda_{\text{closed}}}{\lambda_{\text{open}}}$ [1]

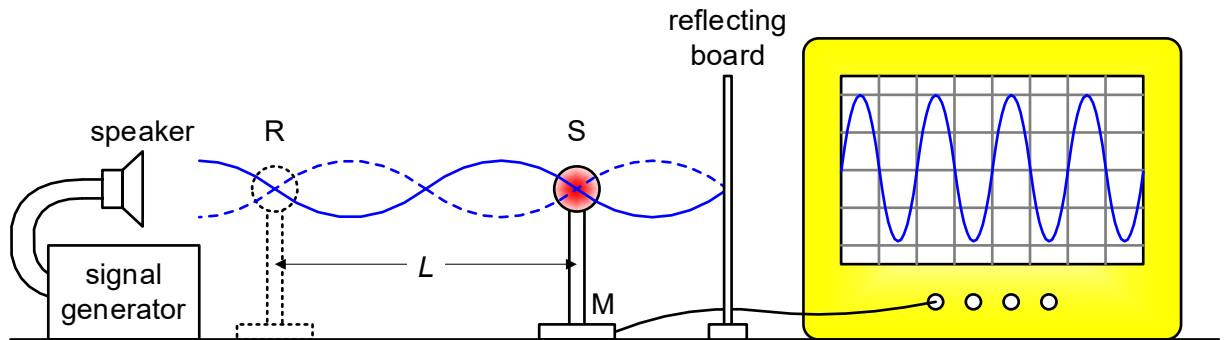
$f_{\text{open}} = f_{\text{closed}} \cdot \frac{\lambda_{\text{closed}}}{\lambda_{\text{open}}} = 512 \times \frac{66.4}{127.2} = 267 \text{ Hz}$ [1]

[N00/2/3 part]

- D3** A loudspeaker S, which is emitting sound of frequency 600 Hz, is placed in front of a reflector R to produce a stationary wave. A sensitive microphone M is moved between R and S, and the signal from M is monitored on a CRO.

CRO signals show that the first peak-signal is detected when M is located at the 130 mm mark on a ruler placed between R and S. The third peak-signal is detected when M is at the 698 mm mark.

Calculate



- (a) the speed of sound in air, and [2]

Wavelength of sound is twice the internodal distance.

$$\lambda = 698 - 130 = 568 \text{ mm}$$

So, speed of sound

$$v = f\lambda = 600 \times 0.568 = 341 \text{ m s}^{-1}$$

- (b) the positions of the two minima that lie between these two peak-signals. [2]

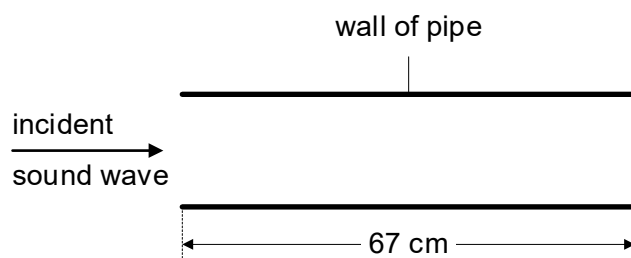
The minimum is located at the midpoint between two maxima. So, their positions are

$$x_1 = 130 + \frac{1}{4}\lambda = 272 \text{ mm} \quad [1]$$

$$x_2 = 130 + \frac{3}{4}\lambda = 556 \text{ mm} \quad [1]$$

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- D4 (a)** A sound wave passes into a pipe that is open at both ends, as shown in the figure below.

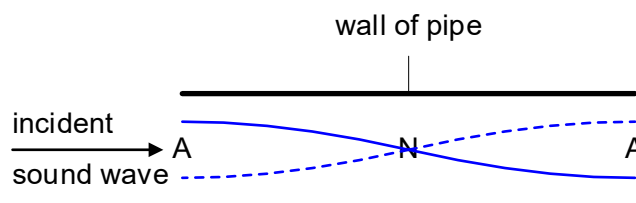


The frequency of the sound wave is gradually increased from a low value. A loud sound is heard in the pipe for the first time at a frequency of 250 Hz. The length of the pipe is 67 cm.

- (i)** On the figure, mark all the positions of

1. the displacement antinodes (use the letter A),
2. the displacement nodes (use the letter N).

[1]



- (ii)** Calculate a value for the speed of sound in the pipe.

[2]

Assuming no end corrections,

$$\text{internodal distance } \frac{1}{2}\lambda = L$$

[1]

Speed of wave

$$v = f\lambda = 250 \times 2 \times 0.67 = 335 \text{ m s}^{-1}$$

[1]

- (b)** An alternative, more reliable, method of measuring the speed of sound shows that the value in **(a)(ii)** is an underestimate.

This underestimate cannot be attributed to the uncertainty in the measurement of either the frequency or the length of the pipe.

State and explain what can be deduced about the positions of either the nodes or the antinodes of the stationary wave in the pipe.

[2]

The antinodes of the stationary wave are located slightly outside the ends of pipe. This is known as the **end correction**.

[1]

The position of the node remains unchanged.

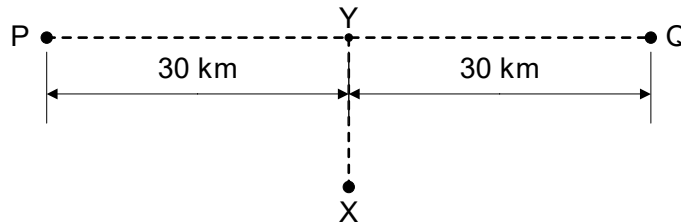
Hence, the **accurate wavelength** of the sound wave $2(L + 2c)$ (c is end correction), instead of $2L$. This leads to an underestimation in the speed of sound ($v_{\text{actual}} = f \times 2(L + 2c)$ vs. $v_{\text{cal}} = f \times 2L$) measured in **(a)(ii)**.

[1]

[N10/3/7 part]

❖ **Two-Source Interference**

- D5** A ship at X is equidistant from two shore-based radio transmitters P and Q as shown below. Both transmitters operate on a wavelength of 300 m and radiate signals of equal amplitude.



- (a) In the figure, the ship at X detects no signal. What information does this give about the signals from P and Q? [1]

The ship detects zero signal implies that the signals from P and Q arrived π rad out of phase.

At X, path difference zero ($PX - QX = 0$). Therefore the signals from P and Q are emitted π rad out of phase. [1]

- (b) The ship moves in a straight line from X to Y. Throughout the journey the amplitude of the signal detected by the ship is zero. Explain this. [2]

The line XY is the perpendicular bisector of line PQ, i.e. all points on XY are equidistant from both P and Q and hence the path difference is always zero. [1]

Since the signals from P and Q are π rad out of phase, they will arrive at any point along XY π rad out of phase to undergo destructive interference. Hence the amplitude of the signal detected is zero (minimum). [1]

- (c) The ship moves in the direction YQ until the signal detected has amplitude twice that from either transmitter alone.

How far has the ship moved? Explain your working. [2]

Between P and Q, a stationary radio wave is generated with Y being a node ($\Delta = 0$).

If the signal detected is twice the amplitude from either transmitter, an antinode ($\Delta = \frac{1}{2}\lambda = 150$ m) is located at this new position. [1]

The path difference between the radio waves arriving at the ship is twice the distance moved along YQ. So, the ship must have moved 75 m. [1]

[The alternative method is to recognise that the distance between adjacent node and antinode is $\frac{1}{4}\lambda$.]

- (d) When the ship sails from Y to the harbour alongside transmitter Q, the detected signal rises and falls in amplitude. Calculate how many dips in intensity will be detected between Y and Q, inclusive. [2]

A node is located whenever a dip in intensity is detected. [1]

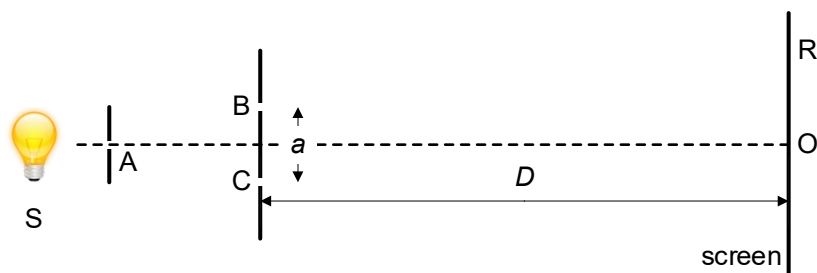
Since distance between two successive nodes is $\frac{1}{2}\lambda = 150$ m, the total number of dips is the number of nodes between Y and Q, i.e., $n = 30000 \div 150 = 200$. [1]

[Maximum path difference at Q is 60 km = 200λ . Since the signals are emitted antiphase, a minimum is detected at Q. So 200 dips in intensity will be detected.]

[Note: Y is a node but it is not considered a dip in intensity since it is the starting position.]

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- D6** The figure below illustrates the Young's double-slit experiment. A source of light S illuminates a narrow slit A, which acts as a source for the narrow slits B and C, and produces fringes on the screen. With light of wavelength λ , bright fringes of separation s are formed on the screen.



- (a) State the relation between λ , s , a , and D . [1]

$$s = \frac{\lambda D}{a} \quad [1]$$

- (b) Suggest suitable values for a and D . [1]

Suitable value of a and D are 0.5 mm and 1 m. [1]

- (c) Describe and explain what happens to the fringes if

- (i) both slits B and C are made narrower whilst keeping d constant, [2]

Narrower slits allow **less light** (less energy) to pass through. [1]

So, the bright fringes are **less bright**. Hence the **contrast** of the fringe pattern is **reduced**. [1]

[No change to fringe separation.]

[Note: Higher order fringes (very dim) may or may not be observed due to greater diffraction by the narrower slits.]

- (ii) the light emerging from slit B is reduced in intensity to half that from slit C, [3]

The **amplitude** of light from slit B ($0.707A$) arriving on the screen is **less than** that from slit C (A). [1]

Resultant amplitude and hence intensity at points of constructive interference is lower, i.e., bright fringes have lower intensities.

Resultant amplitude and hence intensity at points of destructive interference is **not zero due to incomplete destructive interference**, i.e., dark fringes have higher intensities. [1]

Therefore, overall **contrast decreases**. [1]

[No change to fringe separation.]

[Analysis: If amplitude of light from C is A , then amplitude of light from B is $0.707A$.

The resultant amplitude for constructive and destructive interference are $1.707A$ and $0.293A$ (incomplete) respectively.

The intensity of maxima and minima are, therefore, $2.914I_0$ ($4I_0$) and $0.085I_0$ (zero), respectively.]

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- (iii) a thin sheet of transparent plastic is inserted between slit B and slit A, [3]

The wavelength of light decreases when it is in a medium of higher refractive index. [1]

Optical path length (OPL) along ABO increases while OPL along ACO remains unchanged. As a result, the optical path difference (OPD) is not zero at O and the zeroth order maximum is not longer located at O. [1]

Zero optical path difference is obtained above O. So, the fringes are shifted towards R with no change in the fringe separation. [1]

[Optical path length OPL in a medium is nL where n is the refractive index of the medium. At O', even though the path difference is zero, OPD is not zero as $OPL \text{ along ABO} > OPL \text{ along ACO}$.]

- (iv) slits B and C are both covered with sheets of polaroid with their polarising axis aligned, and that in front of B is slowly rotated. [2]

The bright fringes appear dimmer when the axis of polarisation is parallel as the polaroid reduces the intensity of the light passing through it.

As the angle between axes of polarisation increases, the contrast of the fringes decreases due to lower intensities of bright fringes and non-zero intensities of dark fringes (complete cancellation of light waves to produce completely dark fringes is no longer possible). [1]

When the angle between axes of polarisation is 90° , light waves from slits B and C do not interfere, i.e., the fringe pattern disappears. The screen is illuminated by single-slit diffraction pattern. [1]

[Note:

Only light waves of same direction of polarisation can interference with one another. Contrast between bright and dark fringes decrease due to:

1. Only parallel component of light wave from slit B can interference with light wave from slit C, resulting is dimmer bright fringes and brighter dark fringes.

2. The perpendicular component of light wave from slit B does not interfere with light wave from slit C. It brightens up the entire region, reducing the contrast further.]

- (d) When the wavelength of the source is 500 nm, the center of the 120th dark fringe counting from O' lies at R.

Upon replacement of the source by one of unknown wavelength, R is found to be the location of the 90th bright fringe (take O' as the 0th order maximum).

Determine the wavelength of the unknown source. [2]

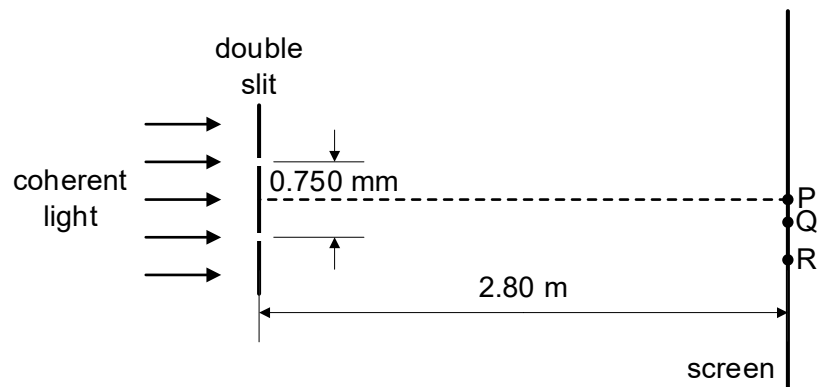
Let the path difference of the light arriving at R from B and C be Δ . Then

$$\Delta = 90\lambda_1 = 119.5\lambda_2 \quad [1]$$

$$\lambda_1 = \frac{119.5\lambda_2}{90} = \frac{119.5 \times 500}{90} = 664 \text{ nm} \quad [1]$$

[J85/3/8 modified]

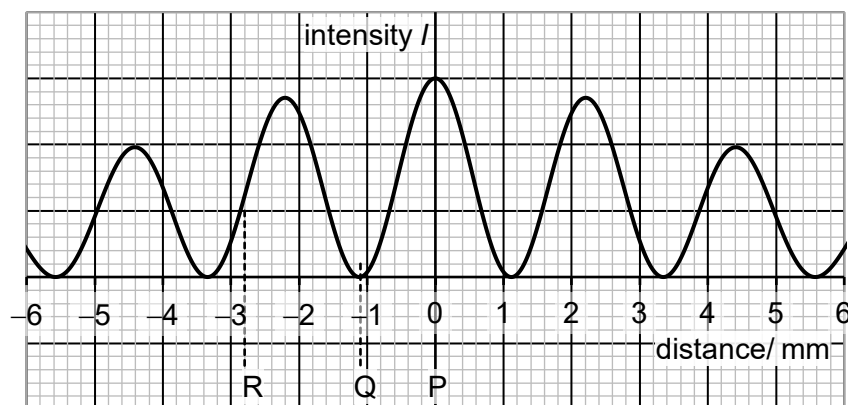
D7 Coherent light is incident normally on a double-slit, as shown in the figure below.



The separation of the slits in the double-slit arrangement is 0.75 mm.

A screen is placed parallel to and at a distance of 2.8 m from the double slit. P is a point on the screen that is equidistant from the two slits.

The figure shows the variation with distance from P of the intensity I of the light on the screen.



(a) Calculate the wavelength, in nm, of the coherent light. [4]

The figure shows that the fringes are equally spaced.

The position of the second order bright fringe is 4.4 mm from P. [1]

Fringe separation

$$x = 4.4 \div 2 = 2.2 \text{ mm} \quad [1]$$

$$\lambda = \frac{xa}{D} = \frac{0.0022 \times 0.75 \times 10^{-3}}{2.8} = 5.89 \times 10^{-7} \text{ m} = 589 \text{ nm} \quad [2]$$

[The fringes are equally spaced if $D \gg a$ and $a \gg \lambda$.]

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- (b) Points Q and R are points on the screen. Their positions are indicated in the graph of intensity variation. Determine the phase angle between the waves from the double slit when the waves meet at

- (i) Point Q [1]

Q is a point of destructive interference. Hence, phase difference is π rad.

- (ii) Point R [2]

Since the fringes are **equally spaced**, we conclude that the phase difference between the waves from the slits must be **directly proportional** to position on the screen, such that every 2.2 mm from P is equivalent to a phase difference of 2π . [1]

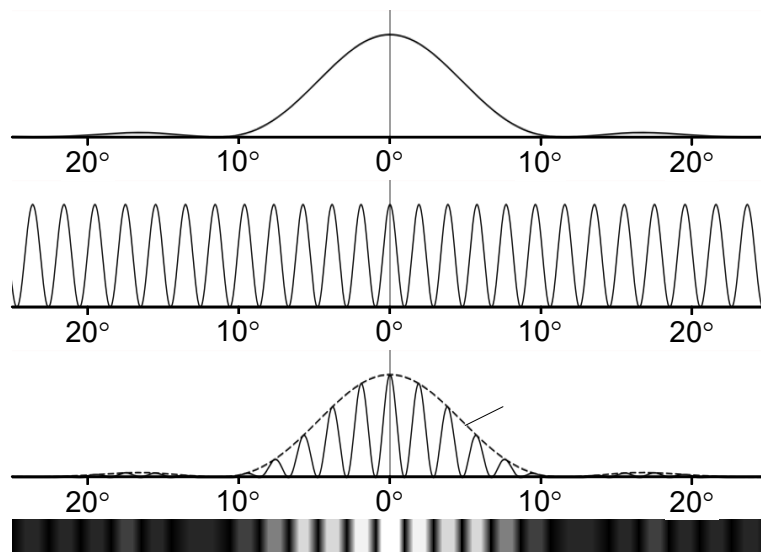
At R,

$$\phi = \frac{0.6}{2.2} \times 2\pi = 0.545\pi$$
 [1]

- (c) Suggest why the maxima on the figure are not all of the same intensity. [2]

Light waves first undergo diffraction as they pass through each individual slit, producing two **single-slit diffraction patterns**. These diffracted waves then **overlap and interfere**, resulting in the interference fringes observed on the screen. [1]

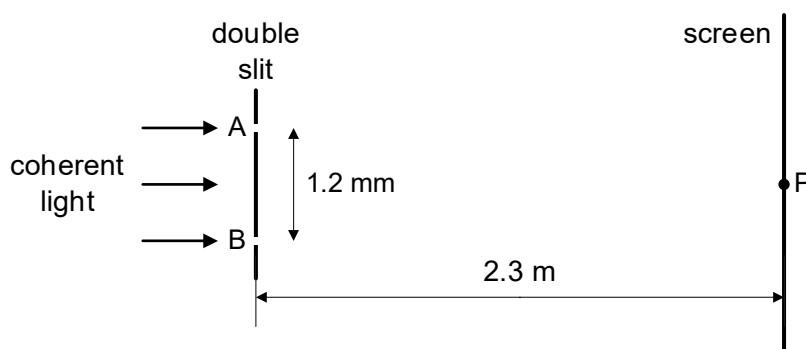
Consequently, the overall interference pattern is **modulated by the broader diffraction envelope** created by each slit. [1]



[N06/2/6]

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- D8** Coherent light of wavelength 590 nm is incident normally on a double slit, as shown in the figure below.



The separation of the slits in the double-slit arrangement is 1.2 mm. Interference fringes are observed on a screen placed parallel to the plane of the double slit and 2.3 m from it. Assume that, for the fringes near point P on the screen, the light reaching the screen from slit A alone has intensity I and that from slit B alone has intensity $\frac{1}{3}I$.

- (a) Determine the separation of the bright fringes. [2]

Fringe separation

$$x = \frac{\lambda D}{a} = \frac{590 \times 10^{-9} \times 2.3}{1.2 \times 10^{-3}} = 1.13 \times 10^{-3} \text{ m} \quad [2]$$

- (b) Point P on the screen is equidistant from the two slits A and B.
Determine the intensity, in terms of I , of a dark fringe near P. [3]

Suppose the amplitude of wave from slit A is A such that $I = kA^2$.

If the intensity of wave from slit B is $\frac{1}{3}I$, then the amplitude of wave from slit B is

$$kA'^2 = \frac{1}{3}I = \frac{1}{3}kA^2$$

$$A' = \frac{1}{\sqrt{3}}A \quad [1]$$

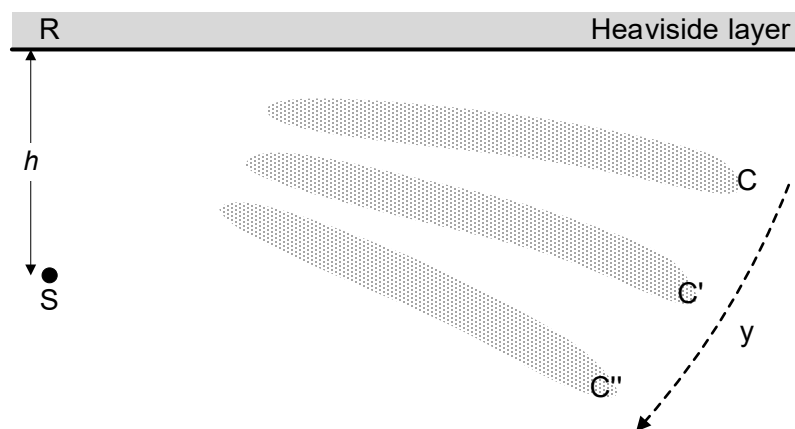
Resultant amplitude of the dark fringe is $A\left(1 - \frac{1}{\sqrt{3}}\right)$. [1]

Intensity of the dark fringe

$$I_{\min} = kA^2 \left(1 - \frac{1}{\sqrt{3}}\right)^2 = 0.179I \quad [1]$$

[N11/3/3 modified]

D9



- (a) A source S of continuous waves a distance h from a plane reflector R produces regions of high intensity such as C, C' and C'' as shown above. Account for this. [2]

The regions of high intensity are the results of **constructive interference** between the **direct wave** and the **reflected wave**. At these positions, the waves meet in phase. [1]

Since the **path difference** ($\Delta = n\lambda$) between the direct and reflected waves **increases** further away from R (from C to C''), the **order of maxima increases** from C to C''. [1]

- (b) When the frequency of S is changed slowly, the regions C, C' and C'' move in the direction y as shown.

Deduce whether the frequency has been increased or decreased. Assume that no phase change occurs upon reflection with R. [2]

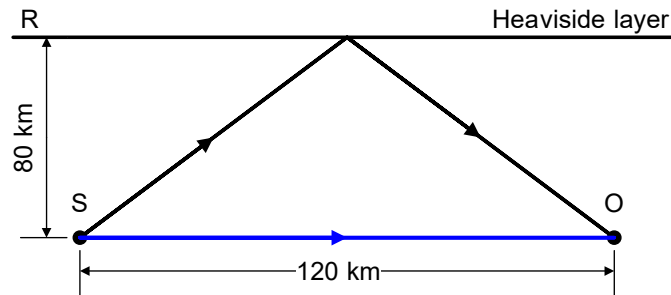
As the regions moves in direction y, the **path difference** Δ at these positions **increases**. [1]

Since the order of maximum of each region remains the **same** (region C is first maximum) and $\Delta = n\lambda$, this is due to **longer wavelength or lower frequency**. [1]

[The order of the maxima remains the same. Hence increasing path difference implies that wavelength has increased since $\Delta = n\lambda$ (assume no phase change during reflection).]

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- (c) In the Appleton's experiment, S was a radio transmitter on the Earth's surface and R was the Heaviside layer – a reflecting layer in the atmosphere 80 km above the ground. When the transmitted wavelength is slowly changed from 200 m to 180 m, a receiver on the ground 120 km away from S observed fluctuations in the received signal strength. Calculate the number of signal maxima observed during this change of frequency. [5]



Path difference

$$\Delta = 2\sqrt{80^2 + 60^2} - 120 = 80 \text{ km} \quad [2]$$

The order of maximum at the two wavelengths (assuming no phase change during reflection) are

$$n_1 = \frac{\Delta}{\lambda_1} = \frac{80000}{200} = 400 \quad [1]$$

$$n_2 = \frac{\Delta}{\lambda_2} = \frac{80000}{180} = 444.4 = 444 \quad [1]$$

The number of signal strength maxima is 44. [1]

[J79/3/1 modified]

❖ **Diffraction Grating**

- D10** A beam of yellow light is shone normally on to a diffraction grating. Bright light is seen emerging at certain angles as shown in Fig. 10(a).

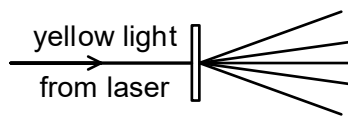


Fig. 10(a)

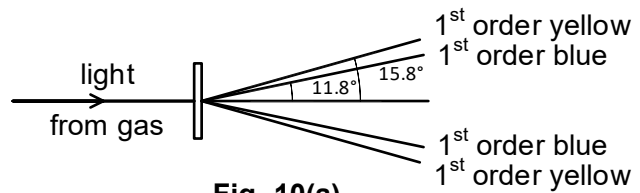


Fig. 10(b)

Use the principle of superposition to suggest a qualitative explanation of this effect. [2]

The laser beam is **diffracted** as it passes through the slits of the grating. These diffracted light waves then interfere in the region after the slits. At certain angles, the **path difference** of the light beams from adjacent slits is **integer multiple of wavelength** of the light. [1]

The beams from all the slits meet in phase and this results in **constructive interference** and bright light is observed at these angles. [1]

- (a) A diffraction grating with a grating spacing of 2.20×10^{-6} m is used to examine the light from a glowing gas. It is found that the first order blue light emerges at an angle 11.8° and the first order yellow light at an angle 15.8° as shown in Fig. 10 (b).

- (i) Calculate the wavelengths of these two colours. [3]

Using $d \sin \theta = n\lambda$,

$$\lambda_b = d \sin \theta_b = 4.50 \times 10^{-7} \text{ m} \quad [2]$$

$$\lambda_y = d \sin \theta_y = 5.99 \times 10^{-7} \text{ m} \quad [1]$$

- (ii) Describe and explain what will be observed at an angle of 54.8° . [3]

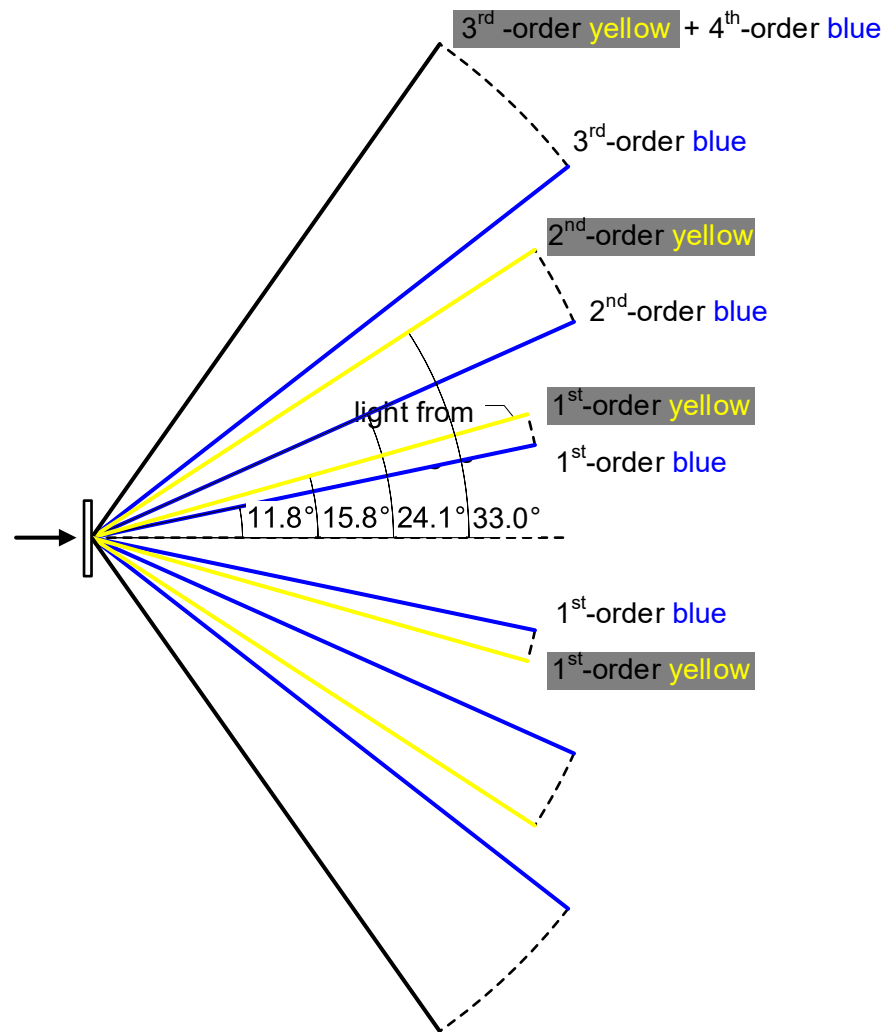
At $\theta = 54.8^\circ$,

$$\Delta = d \sin \theta = 1.80 \times 10^{-6} = 3\lambda_y = 4\lambda_b \quad [2]$$

Hence, the 4th order blue light coincides with 3rd order yellow light. [1]

- (iii) Without making any further calculations, draw a sketch similar to Fig. 10(b) showing the whole pattern observed. [2]

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[J93/3/2 modified]

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D11 A narrow beam of coherent light of wavelength 589 nm is incident normally on a diffraction grating having 4.00×10^5 lines per metre.

- (a) Determine the number of orders of diffracted light that are visible on each side of zero order. [3]

Fringes can be observed for angles $\theta \leq 90^\circ$. [1]

$$n = \frac{a \sin \theta}{\lambda} \leq \frac{a}{\lambda} = \frac{1}{4.00 \times 10^5 \times 589 \times 10^{-9}} = 4.2 \quad [1]$$

Maximum order of bright fringes is 4. [1]

- (b) A student suspects that there are two wavelengths of light in the incident beam, one at 589.0 nm and the other at 589.6 nm.

- (i) State the order of diffracted light at which the two wavelengths are most likely to be distinguished. [1]

4th order.

- (ii) The minimum angular separation of the diffracted light for which two wavelengths may be distinguished is 0.10° . Make calculations to determine whether the student can observe the two wavelengths as separate images. [3]

For the 4th order diffracted light,

$$\theta_A = \sin^{-1} \left(\frac{4\lambda_A}{d} \right) = \sin^{-1} (4 \times 4.00 \times 10^5 \times 589.0 \times 10^{-9}) = 70.46^\circ$$

$$\theta_B = \sin^{-1} \left(\frac{4\lambda_B}{d} \right) = \sin^{-1} (4 \times 4.00 \times 10^5 \times 589.6 \times 10^{-9}) = 70.62^\circ \quad [1]$$

The angular separation

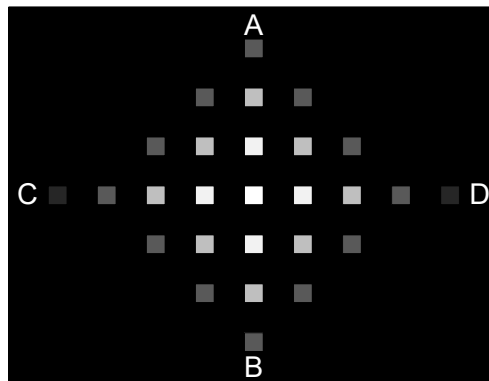
$$\Delta\theta = 0.16^\circ \quad [1]$$

The two bright fringes can be resolved since the angular separation is greater than the angular resolution. [1]

[N03/2/5 part]

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- D12** When a distant streetlight, which is behaving as a point source of monochromatic light of wavelength 5.90×10^{-7} m, passes through a nylon net curtain, the pattern of light incident on a screen is shown in the diagram below.



The main feature of this pattern is the two lines (AB and CD) of bright images.

- (a) State which line of AB or CD is produced by the horizontal nylon threads. [1]

AB [1]

- (b) Determine the number of threads per millimetre in the nylon net curtain if the angle as viewed between the central image and the first adjacent image is 0.60° ? [4]

Thread separation

$$d = \frac{\lambda}{\sin \theta} = \frac{5.90 \times 10^{-7}}{\sin 0.60^\circ} = 5.63 \times 10^{-5} \text{ m}$$

Number of threads is

$$N = \frac{1}{d} = \frac{1}{5.63 \times 10^{-5}} = 17700 \text{ m}^{-1} = 17.7 \text{ mm}^{-1} \quad [2]$$

- (c) Why is the spacing of the images constant? [1]

The spacing of the images constant because the thread separation is much larger (95x) compared to the wavelength of light. [1]

[N00/2/4 modified]

❖ **Single-Slit Diffraction and Rayleigh Criterion**

- D13** Parallel light of wavelength 590 nm is incident on a rectangular slit of width 0.60 mm, as shown in Fig. 13.1.

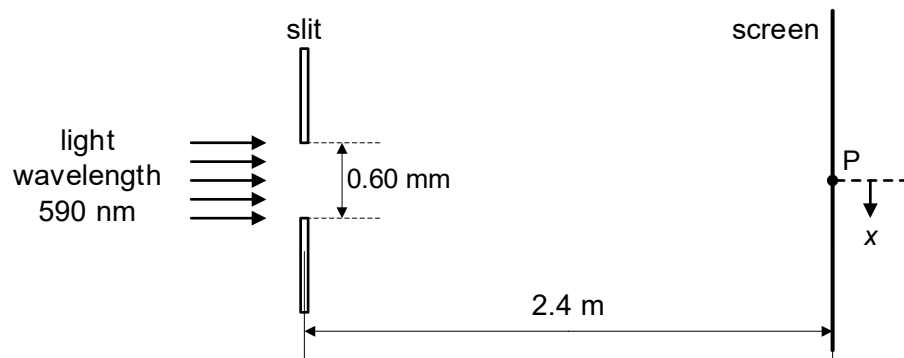


Fig. 13.1

Light passing through the slit is incident on a screen that is 2.4 m from the plane of the slit. The centre of the interference pattern formed on the screen is at P.

- (a) (i) Calculate the width of the central fringe, as observed on the screen. [3]

Angular position of minima

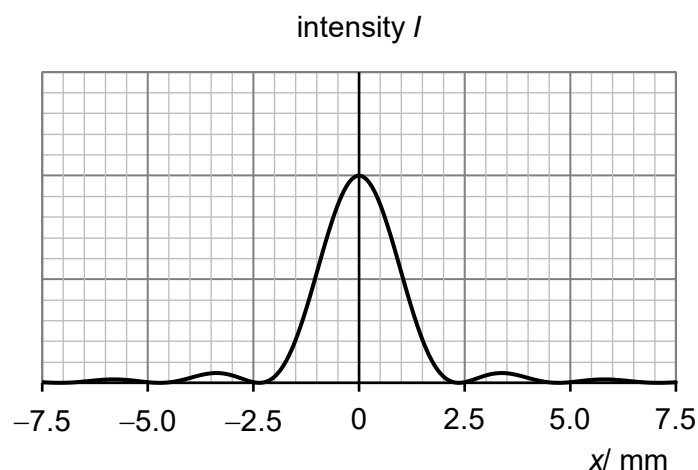
$$\sin \theta = \frac{\lambda}{b} = \frac{590 \times 10^{-9}}{0.60 \times 10^{-3}} = 9.83 \times 10^{-4} \quad [1]$$

Since θ is small, $\sin \theta = \theta$. [1]

Width of central bright fringe

$$\text{width} = D(2\theta) = 2.4 \times (2 \times 9.83 \times 10^{-4}) = 4.72 \times 10^{-3} \text{ m} \quad [1]$$

- (ii) On Fig. 13.2, sketch a graph to show the variation with distance x from point P of the intensity I of the light on the screen. [3]



At least second orders secondary maxima. [1]

Central bright fringe is

- twice as wide as secondary bright fringes. [1]
- at least 10x high intensity as secondary bright fringes. [1]

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YEAR 56 PHYSICS DEPARTMENT**

- (b) Parallel light from a second source and of the same wavelength 590 nm is also incident on the slit. The angle between the two beams of light is θ , as shown in Fig. 13.3.

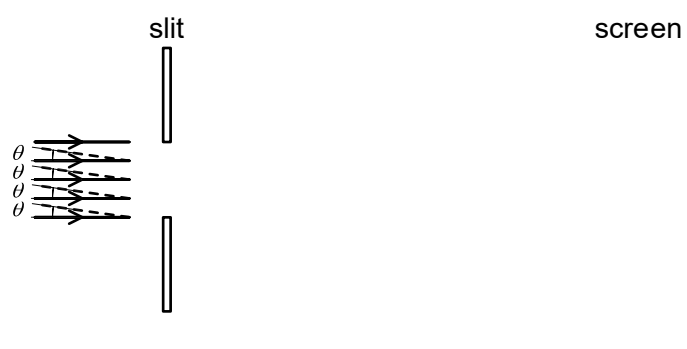


Fig. 13.3

Each beam forms a separate interference pattern on the screen.

- (i) Explain what is meant by the Rayleigh criterion for the resolution of the two patterns. [2]

The two diffraction patterns are **just resolved** [1] when the **central maximum** of one pattern **coincides** with the **first minimum** of the other pattern. [1]

- (ii) Calculate the angle θ , in rad, such that the two interference patterns are just resolved. [2]

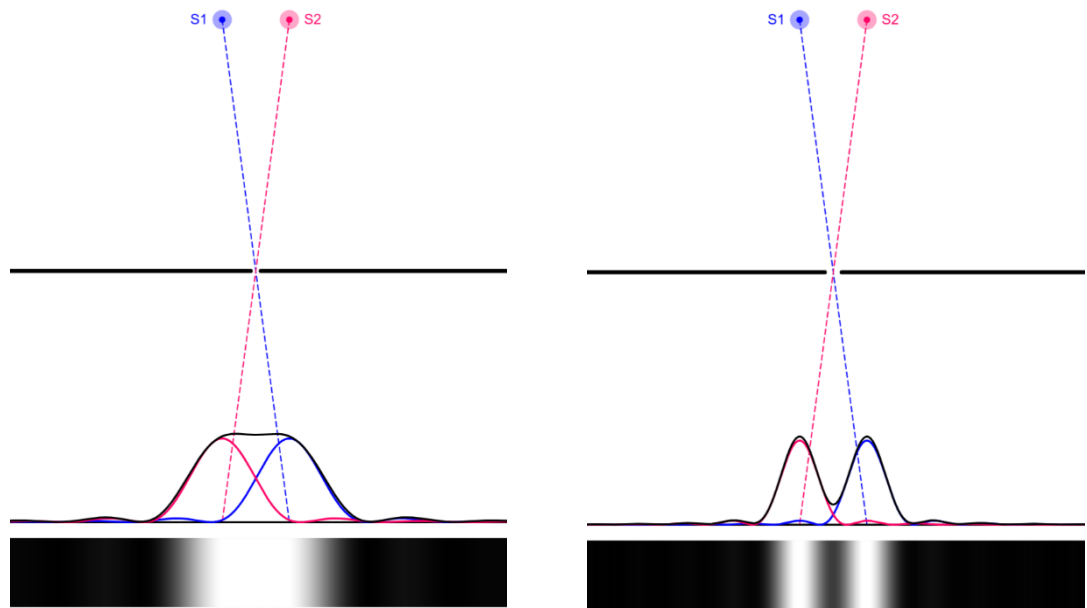
Minimum angular separation

$$\theta_{\min} \approx \frac{\lambda}{b} = \frac{590 \times 10^{-9}}{0.60 \times 10^{-3}} = 9.83 \times 10^{-4} \text{ rad} \quad [2]$$

[N17/3/5]

D14 Two closely spaced monochromatic point sources of light are viewed through a single slit of width w .

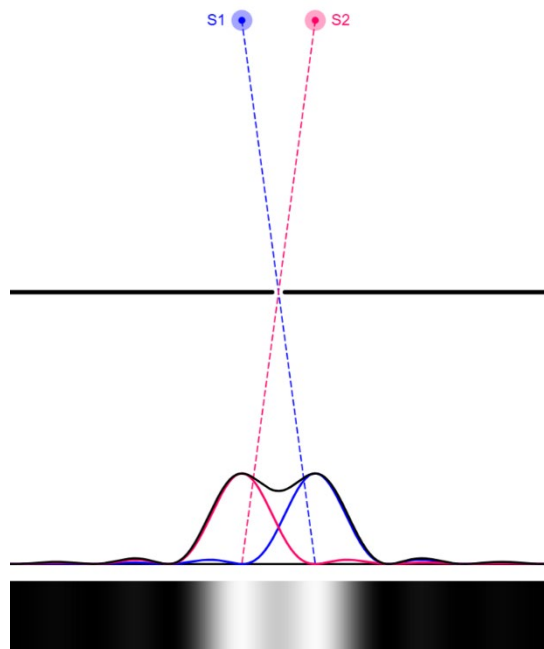
(a) Sketch the appearance of the images observed for small and large values of w . [2]

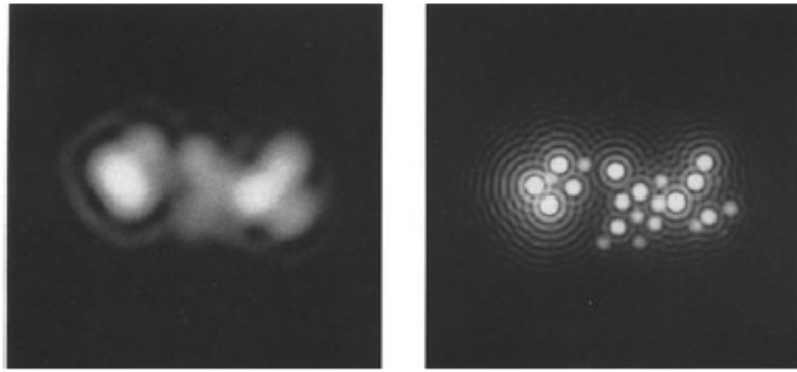


small value of w

large value of w

(b) Sketch a graph showing the intensity variations along a line through the centres of the images for a critical value of w when the images are just resolvable. [2]





Point sources of light viewed through a circular aperture.

- (c) Suggest a way to improve the resolving power of an imaging system which is used to observe small objects or small separation between objects. [1]

A higher resolving power requires the **width of central maximum to be narrower**, i.e., angular position of 1st minimum to be smaller.

This can be achieved with light of **shorter wavelength or aperture of larger size**. [1]

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D15 A beam of monochromatic light of wavelength 600 nm is incident on a single slit of width 0.0200 mm.

- (a) Calculate the resolving power of the slit. [2]

Resolving power of slit

$$\theta_{\min} = \sin^{-1} \frac{\lambda}{b} = \sin^{-1} \frac{600 \times 10^{-9}}{0.0200 \times 10^{-3}} = 0.0300 \text{ rad} \quad [2]$$

- (b) What is the minimum separation between two small objects placed 1.00 km from the slit, in order to be resolved by the slit? [2]

Minimum separation

$$s_{\min} = r \theta_{\min} = 1000 \times 0.0300 = 30.0 \text{ m} \quad [2]$$

- (c) Suppose the single slit is now brought into a medium of refractive index $n = 1.33$.

What would be the answer for (b)? [3]

Wavelength in water

$$\lambda' = \frac{\lambda}{n} = 451 \times 10^{-9} \text{ m} \quad [1]$$

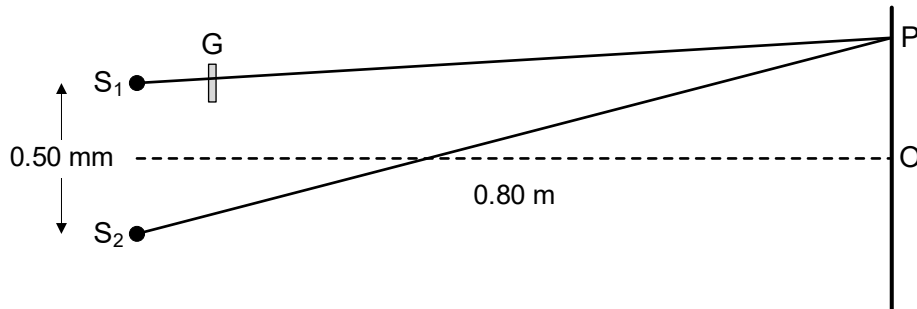
So minimum separation in water

$$s'_{\min} = r \sin^{-1} \frac{\lambda'}{b} = 1000 \sin^{-1} \frac{4.51 \times 10^{-9}}{0.0200 \times 10^{-3}} = 22.6 \text{ m} \quad [2]$$

[Refractive index n of a medium is the ratio of the speed of light in vacuum to that in the medium.]

1.2 Challenging Questions

- C1** In the figure below, S_1 and S_2 are two coherent monochromatic light sources separated by a distance of 0.50 mm. Position O is equidistant from S_1 and S_2 , and the screen is 0.80 m from the slits.



When a thin parallel-sided piece of glass G of thickness 3.6×10^{-6} m is placed in front of S_1 as shown, the centre of the fringe system moves from O to a point P.

Calculate OP, if the wavelength of the two sources is 6.0×10^{-7} m and the refractive index of the glass is 1.5.

Answer: 2.88×10^{-3} m

Optical path difference

$$\Delta_{\text{opd}} = nt - t = m\lambda$$

$$m = \frac{nt - t}{\lambda} = \frac{3.6 \times 10^{-6} \times (1.5 - 1.0)}{6.0 \times 10^{-7}} = 3$$

3rd order bright fringes is now located at O, implying that the fringe pattern is shifted towards P by 3 fringe separation.

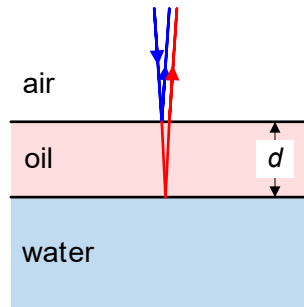
$$OP = \frac{3\lambda D}{a} = \frac{3 \times 6.0 \times 10^{-7} \times 0.80}{0.50 \times 10^{-3}} = 2.88 \times 10^{-3} \text{ m}$$

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- C2** An oil film floating in water is illuminated by white light at normal incidence. The film is 280 nm thick and has a refractive index of 1.45. By considering interference of light rays which are reflected at the air-oil and oil-water boundary, find the dominant colour in the reflected light. Take the refractive indices of air and water to be 1.00 and 1.30, respectively.

[Hint: When light enters a denser medium from a less-dense medium, it will undergo a 180° phase shift. No such phase shift occurs when light enters an optically less dense medium from a denser one.]

Answer: 541 nm (green)



Light is reflected at both the air-oil interface and the oil-water interfaces. Light undergoes 180° phase change upon reflection at the air-oil interface ($n_{\text{air}} < n_{\text{oil}}$) and no phase change at the oil-water interface ($n_{\text{oil}} > n_{\text{water}}$).

The dominant colour of the reflected light corresponds to constructive interference of certain wavelengths.

Condition for constructive interference:

$$\text{OPD} = n_{\text{oil}} \times 2d = \left(m + \frac{1}{2}\right) \lambda$$

Rearranging,

$$\lambda = \frac{2n_{\text{oil}}d}{m + \frac{1}{2}}$$

m	λ / nm
0	1624 (IR)
1	541 (green)
2	325 (UV)

Green light is strongly reflected as IR and UV are not visible.

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1.3 Numerical Answers

D2 (a) 1.8 cm (b) 267 Hz

D3 (a) 341 m s^{-1} (b) 272 mm, 556 mm

D4 335 m s^{-1}

D5 (c) 75 m (d) 200

D6 664 nm

D7 (a) 589 nm (b)(i) 3.14 rad (ii) 1.7 rad

D8 (a) $1.13 \times 10^{-3} \text{ m}$ (b) $0.179 I$

D9 44

D10 (b)(i) $4.50 \times 10^{-7} \text{ m}$, $5.99 \times 10^{-7} \text{ m}$

D11 4

D12 17.7 lines per mm

D13 (a)(i) $4.72 \times 10^{-3} \text{ m}$ (b)(ii) $9.83 \times 10^{-4} \text{ rad}$

D15 (a) 0.0300 rad (b) 30.0 m (c) 22.6 m